

Creative-destruction in the AI industry

Paul Delbouve

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Developing frontier artificial-intelligence models is becoming increasingly expensive¹ (Cottier et al., 2024). Training and deploying the most capable systems requires growing amounts of computing power, specialised labour, data and energy.² Therefore, leading AI developers are committing ever larger sums to data centres, chips and model development (Sha et al., 2026). These investments are risky: their costs are incurred upfront, while their technological and commercial success remains by definition, uncertain. This raises a familiar issue in the economics of innovation. Firms invest only if successful innovation can generate a sufficient return. A technological advance therefore needs to create a temporary advantage during which the innovator can charge higher prices, attract new users and/or gain market share (Aghion et al., 2014). These rents are thus part of the incentive to undertake costly and uncertain research in the first place (Aghion et al., 2015). If competitors can reproduce an innovation immediately, the expected return may become too small to justify the initial investment (Bloom et al., 2013).

The AI sector may be one of best examples of this innovation dilemma. Recent months have seen the emergence of increasingly capable, lower-cost models. Providers such as Deepseek, Qwen or Z-AI sometimes manage to approach the performance of frontier models within a few months while charging much lower prices (Figure 1) : for a given quality level, prices can fall by as much as 90% in less than a year. Several characteristics of AI may make catch-up unusually rapid. Scientific results and engineering advances circulate through research papers, technical reports and open-source software.³ Developers can build on common libraries, publicly available models and a shared stock of knowledge rather than reproduce every stage of the original discovery process (Azoulay et al., 2024). Techniques such as distillation can also allow smaller models to learn from the outputs or reasoning patterns of more capable systems⁴ (Hinton et al., 2015; Mukherjee et al., 2023). Once a technological path has been demonstrated, approaching it may therefore be substantially less costly than opening it in the first place (Guo et al., 2025).

¹Researchers from Epoch.AI specialized in AI cost evaluation estimate that the amortized hardware and energy costs of frontier training runs grew by approximately 2.4 times per year between 2016 and 2024.

²The International Energy Agency reports that global investment in data centres reached approximately \$500 billion in 2024 driven mainly by AI demand

³Hugging-Face is the most used website on which cutting edge models are in open access. It gathers more than 1.3M models and measured 46B models downloads in late 2025.

⁴In February 2026, Anthropic accused Chinese developers DeepSeek, Moonshot, and MiniMax of industrial-scale distillation campaigns. Although these findings do not isolate the full contribution of distillation to frontier catch-up, they support the view that access to high-quality models can lower the cost of catching up with them.

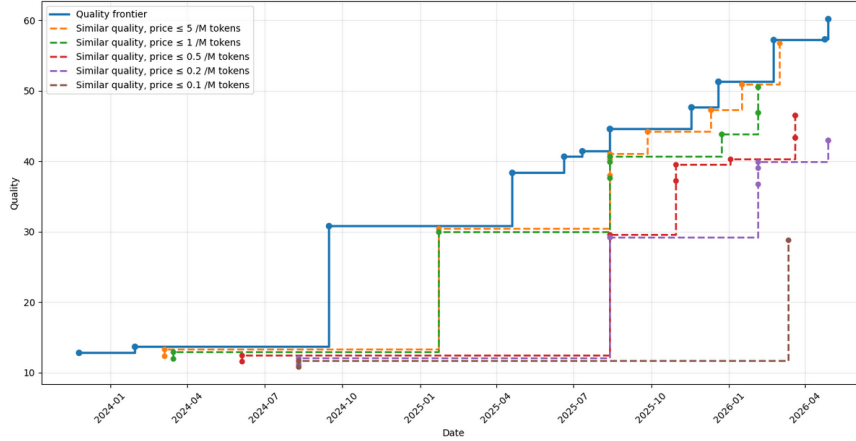


Figure 1: Re-catch time for lower price models (*Data : Artificial Analysis*)

This creates a tension for innovation policy. On the one hand, rapid catch-up benefits users and AI diffusion within the economy. It lowers prices, widens access to advanced capabilities and limits the ability of a small number of firms to maintain dominant positions. It can also put pressure on frontier providers to innovate again in order to preserve their technological lead resulting in a positive effect on AI development (Aghion et al., 2005; Akcigit and Ates, 2019). On the other hand, if catching up is too easy and too fast, it may shorten the period during which frontier developers can recover their investments. A firm that spends heavily to push the technological frontier may see its market share and pricing power eroded before it has earned a sufficient return. In that case, competition would not simply redistribute demand toward cheaper suppliers, it could eventually weaken the incentive to undertake the next costly frontier advance (Aghion et al., 2001; Bloom et al., 2013).

Given that rapid catch-up appears difficult to avoid because of the characteristics of AI technologies, the central question is whether developing frontier models generates enough additional demand to compensate providers for the rapid erosion of their relative advantage. This note therefore estimates the magnitude and persistence of the resulting gains in usage, revenue, and market share, together with the scale and pace of their erosion by lower-cost followers.

Methodology

To examine these mechanisms, we combine weekly data from [OpenRouter](#), a platform that provides access to hundreds of AI models through a single interface, with independent measures of model capabilities from [Artificial Analysis](#). The resulting dataset follows more than 470 models from 82 providers between November 2023 and June 2026 and records their token usage, posted prices, provider identity and measured quality. This high-frequency information allows us to observe both the release of models that push the technological frontier and the subsequent entry of cheaper models offering comparable capabilities. We then estimate how these events affect the usage, revenues and market shares of both frontier providers and exposed incumbent models. OpenRouter represents only one distribution channel and does not cover the entire global AI market; the analysis is therefore

better suited to measuring market reallocation and competitive pressure than total industry revenues.

Main results

The first result is the exceptional expansion of the market. Weekly usage on OpenRouter rose from around 8 billion tokens in November 2023 to more than 41 trillion by June 2026, following an increasingly steep, near-exponential trajectory. This acceleration is not evenly distributed across models: recent growth is concentrated among the most capable systems, consistent with the idea that improvements in quality unlock new applications and increase the range of tasks for which AI can be used economically.

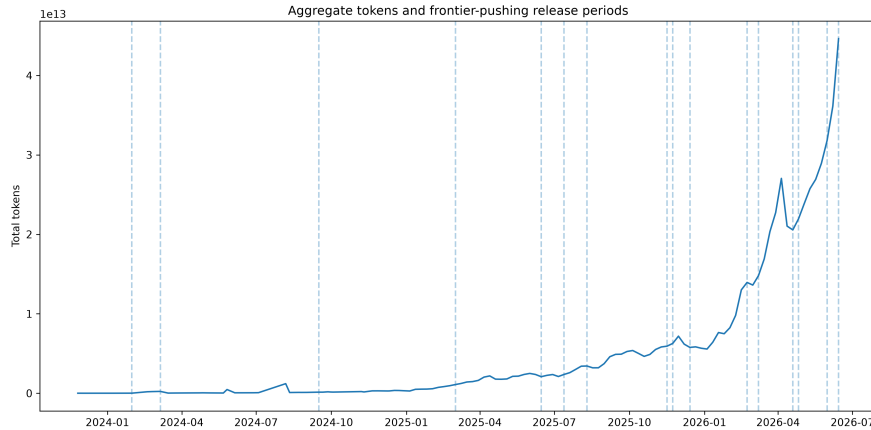


Figure 2: Evolution of weekly tokens sold with frontier increase events

The market is also structured around relatively distinct provider profiles. A small group of firms (OpenAI, Anthropic and Google) repeatedly releases frontier models and generally charges premium prices. Other providers, such as DeepSeek and Moonshot, specialize in approaching recently achieved levels of capability at substantially lower prices. Between these groups are intermediate-price followers and providers whose models remain further from the frontier. Technological catch-up is therefore not simply the result of leading firms differentiating their own product ranges: it frequently comes from competing providers pursuing a lower-cost strategy.

This competition has produced a visible reallocation of token usage. As Figure 4 shows, lower-cost followers have gained token market share, while concentration in terms of tokens has declined since its peak in early 2025. The reallocation is much less pronounced in terms of revenue, however. Frontier providers continue to charge substantially higher prices and therefore remain dominant in market expenditure, even as their share of tokens is eroded. A leading provider can consequently lose part of its usage share while continuing to record rapidly rising revenues.

Our econometric results confirm these two opposing mechanisms. As summarized in Table 2 and Figure 6 when a provider releases a model that pushes the quality frontier, its token usage increases by approximately 0.43 log points and its imputed revenue by 0.70 log points while its gains in market share (Figure 6b) appear smaller and more temporary. Conversely Table 1 shows that, when a cheaper model enters within a comparable quality segment, the usage of exposed incumbent models falls by

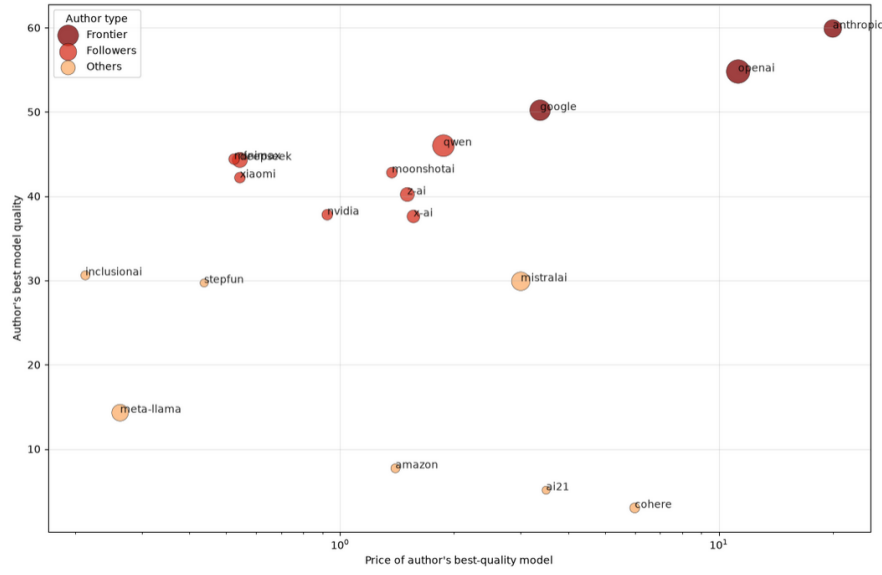


Figure 3: Price of each provider's highest-quality model

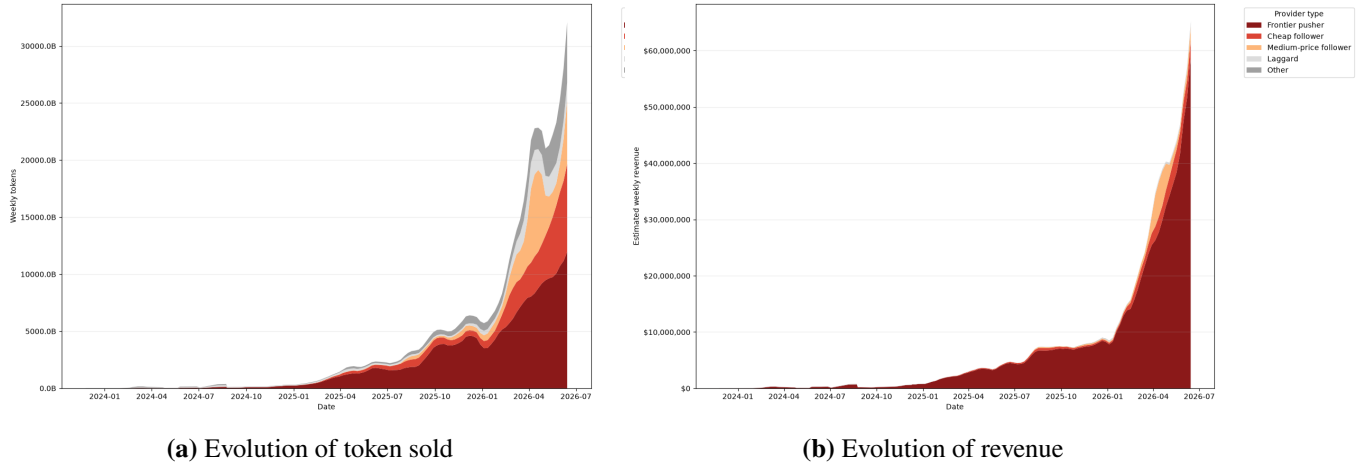


Figure 4: Evolution of tokens and revenue by provider groups

around 0.53 log points, while their market share declines by around 0.53 to 1.9 percentage points depending on token or revenue measurement. These estimates provide direct evidence of business stealing: lower-cost catch-up diverts demand and expenditure away from previously established models.

Taken together, the results highlight a process of creative destruction occurring within a rapidly expanding market. Low-cost followers erode the relative rents of frontier developers by taking market share from them. But this erosion has so far been more than offset by the growth of aggregate demand, particularly for high-capability models. Frontier providers therefore continue to expand in absolute terms even as their competitive advantage becomes less durable. The sustainability of this configuration, however, appears to depend on market expansion continuing to outpace the loss of token market share.

Conclusion

This note shows that lower-cost catch-up creates genuine market erosion. When a cheaper model of comparable quality enters, incumbent models lose usage and revenue market share, while lower-cost providers have secured a growing and increasingly persistent position in the market. So far, however, this competitive pressure has been offset by rapid market expansion. Frontier releases still generate large and sustained gains in usage and, especially, in revenue. Frontier providers can therefore continue to grow in absolute terms even as their usage market shares erode.

The sustainability of this equilibrium remains uncertain. It depends first on AI adoption and demand continuing to expand sufficiently quickly. If market growth slows, stalls or begins to decelerate, losses in market share may translate more directly into weaker revenues and lower innovation rents. Furthermore, it depends on frontier providers remaining able to create recurrent quality improvements. If progress based on current large-language-model architectures becomes more difficult, frontier providers may struggle to maintain a sufficient quality gap over cheaper followers. Rapid catch-up therefore does not yet appear to have made frontier innovation unsustainable. The sustainability of this configuration, however, appears to depend on market expansion continuing to offset losses in token market share.

Appendix

Effect of a price decline

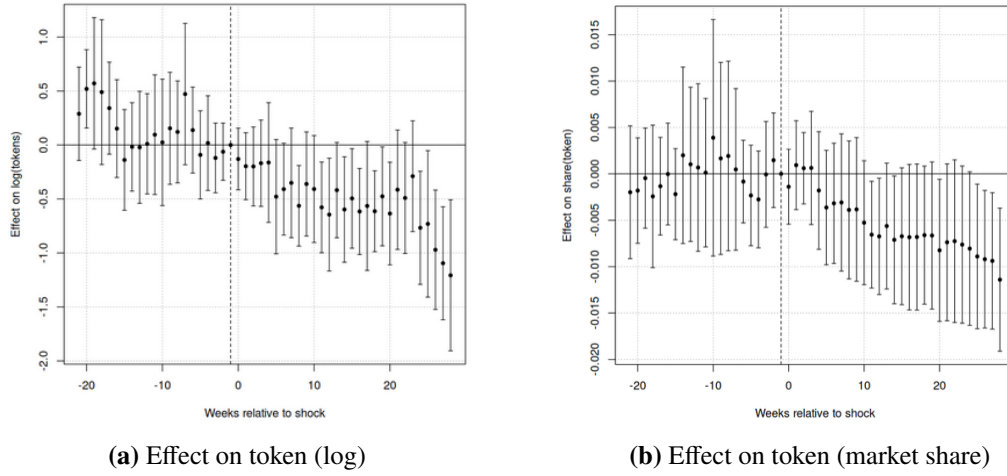


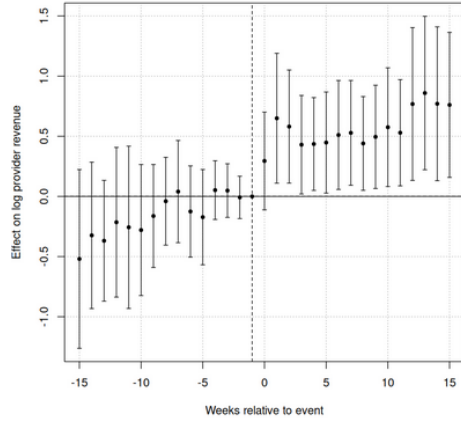
Figure 5: Effect of a price decline on incumbent models token

	Log tokens	Token market share	Revenue market share
	−0.5313** (0.1475)	−0.0053 [†] (0.0026)	−0.0188** (0.0061)
Observations	15,196	30,777	30,777
Adjusted R^2	0.893	0.477	0.547
Within R^2	0.012	0.002	0.006
Model–shock fixed effects	Yes	Yes	Yes
Shock–calendar fixed effects	Yes	Yes	Yes

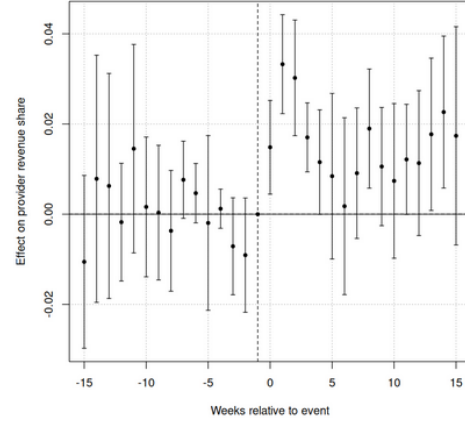
Notes: [†] $p < 0.10$, $*p < 0.05$, $**p < 0.01$, and $***p < 0.001$.

Table 1: Effect of lower-price model entry on incumbent outcomes

Effect of frontier model release



(a) Effect on provider's revenue (log)



(b) Effect on provider's revenue (market share)

Figure 6: Effect of publishing a frontier model on provider's sales in log and market share

	Revenue share	Token share	Log revenue	Log tokens
	0.0149** (0.0052)	0.0137 [†] (0.0072)	0.6993** (0.2504)	0.4318* (0.1854)
Observations	25,144	25,144	21,031	21,239
Adjusted R^2	0.948	0.736	0.882	0.889
Within R^2	0.002	0.002	0.008	0.005
Provider–event fixed effects	Yes	Yes	Yes	Yes
Event-time–event fixed effects	Yes	Yes	Yes	Yes

Notes: OLS estimates. Standard errors, reported in parentheses, are two-way clustered by event and provider. Revenue and token market shares are expressed as fractions between zero and one. The dependent variables in columns 3 and 4 are the logarithms of provider revenue and provider tokens, respectively. [†] $p < 0.10$, $*p < 0.05$, $**p < 0.01$, and $***p < 0.001$.

Table 2: Effect of publishing a frontier model

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